

Experiment 10

Aim: Write a python program to perform Named Entity Recognition using NLTK.

Theory:

Named Entity Recognition (NER) is a Natural Language Processing technique used to identify and classify important entities in text into predefined categories such as person names, locations, organizations, dates, time, and other proper nouns. It helps convert unstructured text into structured information by detecting meaningful elements within a sentence.

In NLTK, Named Entity Recognition is performed using multiple steps. First, the input text is tokenized into sentences and words. Next, Part-of-Speech (POS) tagging is applied to assign grammatical labels such as noun, verb, adjective, etc. After POS tagging, NLTK uses the `ne_chunk()` function to group words into named entity chunks. This function uses a pre-trained model to identify entities and classify them into categories like PERSON, ORGANIZATION, GPE (Geopolitical Entity), LOCATION, and others.

The output of NER is typically displayed as a tree structure, where identified entities are grouped and labeled accordingly. This process helps in automatically extracting important information such as names of people, places, institutions, and dates from large text documents.

Named Entity Recognition is widely used in applications such as information extraction, search engines, document summarization, question answering systems, chatbots, and content analysis, where identifying key entities improves understanding of text data.

Source Code and Outputs:

```
!pip install nltk

import nltk
from nltk.tokenize import word_tokenize
from nltk import pos_tag, ne_chunk

... Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex<=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.3)

nltk.download('punkt')
nltk.download('maxent_ne_chunker_tab')
nltk.download('punkt_tab')
nltk.download('averaged_perceptron_tagger')
nltk.download('averaged_perceptron_tagger_eng')
nltk.download('maxent_ne_chunker')
nltk.download('words')

[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package maxent_ne_chunker_tab to
[nltk_data] /root/nltk_data...
[nltk_data] Unzipping chunkers/maxent_ne_chunker_tab.zip.
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data] /root/nltk_data...
[nltk_data] Package averaged_perceptron_tagger is already up-to-
[nltk_data] date!
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data] /root/nltk_data...
[nltk_data] Package averaged_perceptron_tagger_eng is already up-to-
[nltk_data] date!
```

```
text = "Elon Musk is the CEO of Tesla and he lives in the United States"

tokens = word_tokenize(text)

print("Tokens:", tokens)

Tokens: ['Elon', 'Musk', 'is', 'the', 'CEO', 'of', 'Tesla', 'and', 'he', 'lives', 'in', 'the', 'United', 'States']

pos_tags = pos_tag(tokens)

print("POS Tags:")
print(pos_tags)

POS Tags:
[('Elon', 'NNP'), ('Musk', 'NNP'), ('is', 'VBZ'), ('the', 'DT'), ('CEO', 'NN'), ('of', 'IN'), ('Tesla', 'NNP'), ('and', 'CC'), ('he', 'PRP'), ('lives',
```

```
ner_tree = ne_chunk(pos_tags)

print("Named Entity Tree:")
print(ner_tree)

Named Entity Tree:
(S
  (PERSON Elon/NNP)
  (ORGANIZATION Musk/NNP)
  is/VBZ
  the/DT
  (ORGANIZATION CEO/NN of/IN Tesla/NNP)
  and/CC
  he/PRP
  lives/VBZ
  in/IN
  the/DT
  (GPE United/NNP States/NNPS))
```

```
from nltk.tree import Tree

def print_tree(tree, indent=0):
    if isinstance(tree, Tree):
        print(' ' * indent + str(tree.label()))
        for child in tree:
            print_tree(child, indent + 1)
    else:
        print(' ' * indent + str(tree))

print_tree(ner_tree)

... S
  PERSON
    ('Elon', 'NNP')
  ORGANIZATION
    ('Musk', 'NNP')
  ('is', 'VBZ')
  ('the', 'DT')
  ORGANIZATION
    ('CEO', 'NN')
    ('of', 'IN')
    ('Tesla', 'NNP')
  ('and', 'CC')
  ('he', 'PRP')
  ('lives', 'VBZ')
  ('in', 'IN')
  ('the', 'DT')
  GPE
    ('United', 'NNP')
    ('States', 'NNPS')
```

Learning Outcomes:

I am able to write Python programs using NLTK to perform Named Entity Recognition by detecting, extracting, and classifying entities such as names of persons, organizations, locations, dates, and more, enabling structured information extraction from unstructured text for various intelligent applications.